PERSONALIZED LEARNING TRACKS (TRACK 3)

SDG PROJECT REPORT

ON

TOURISM PROMOTION WEBSITE

Submitted By

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To

the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

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(Approved by AICTE and affiliated to APJ Abdul Kalam Technological University)

AUGUST 2026

SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)

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BONAFIDE CERTIFICATE

Certified that the report entitled **"Tourism Promotion Website"** submitted by **"SOJU YOHANNAN - MGP25MCA-2058"** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of **Master of Computer Applications** is a bonafide record of the **Social Immersion Sustainable Development Goals(SDG)** project work carried out by him / her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

At the outset, I thank the Lord Almighty for His abundant grace, strength and hope to make our endeavour a success. I express my deep-felt gratitude to **Dr. Sudha T**, Principal, Saintgits College of Engineering (Autonomous), for her warm support with regard to the work and to the management for providing the facilities required.

I would like to place my deep sense of gratitude to **Prof. Mini Punnoose**, Director MCA, Department of Computer Applications, Saintgits College of Engineering (Autonomous), for her constant encouragement. I express my gratitude to **Dr. Rani Saritha R**, Head of the Department, Department of Computer Applications, Saintgits College of Engineering (Autonomous), for her support and guidance.

I am profoundly grateful to **Dr.Amrita Priya K**, my SDG Project Guide and **Ms. Ankitha Philip**, the SDG Project Coordinator, for their valuable guidance, help, suggestions and assessment.

Furthermore, I would like to thank all others, especially my parents and numerous friends. This project report would not have been a success without their inspiration, valuable suggestions and moral support from them throughout its course.

Soju Yohannan

ABSTRACT

The Tourism Promotion Website is a multifunctional online platform created to rescue the local economy and help the community. The website offers an interactive interface for travelers who want to explore the place and sign up for local packages. Moreover, it enables the local service providers to promote their business and attract clients from all over the world. The application is con- structed using the Node.js and Express framework on the server side and utilizes MySQL relational database to design the data schema. It implements the JWT (JSON Web Token) authentication mechanism to ensure secure authorization be- tween the users and the server. The website has three major roles: Admin, Tourist, and Service Provider. An administrator controls the entire system, moderates the services provided by local businessmen, and manages destinations and travel packages. Service providers have access to the dashboard, where they can post ad- vertisements and monitor the tourists' feedback. Finally, the tourists interact with the website by searching for destinations, reviewing and booking their desired travel packages and services. The Tourism Promotion Website can also be used to advertise local prod- ucts and introduce travelers to the local culture and unique places that are off the beaten path. Therefore, the application helps increase the overall economic impact of tourism on the local economy and creates sustainable employment opportuni- ties for the community members. It empowers the local small businesses and re- staurants by connecting them with the tourists directly. In addition, it promot- es cultural diversity and sustainable tourism. Hence, the project aligns with the United Nations' SDGs, specifically the 8th Sustainable Development Goal, decent work, and economic growth, by creating job opportunities for the locals. Overall, the application uses modern technologies to connect tourists with the local com- munity, provide the necessary infrastructure for economic growth, and empower the businessmen by making their services available online.

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

The **Tourism Promotion Website** is a web application that was designed to establish connections between tourists and local service providers in India's popular destinations. The website functions as a marketplace, offering a space where guests can discover unique heritage experiences, reserve travel packages and provide feedback. At the same time, tourism-related businesses can use the platform to promote their brands, list packages and services and interact with clients.

The application was developed within the framework of the Social Immersion Sustainable Development Goals (SDGs) through Personalized Learning Tracks (Track 3). It aims to facilitate social and economic sustainability of local destinations through an e-commerce platform.

There are three primary user categories in the Tourism Promotion Web Application:

- **Administrator:** Adds destinations and accommodations, moderates users, approves providers, and manages other aspects of the site.
- **Tourist:** Explores the website, browses destinations, books travel packages, accesses services and leaves feedback.
- **Service Provider:** Registers on the website to set up a business catalog and promote tourism-related services.

The application was built with **Node.js** and **Express.js** framework with **MySQL** as a database. It was developed with **HTML5, CSS3, JavaScript** and **Bootstrap 5** as front-end technologies. The authentication system is handled by **JSON Web Tokens (JWT)** and server-side sessions.

1.2 Objectives

The main objectives of this project are:

1. To build a web-based marketplace where users can discover destinations and book travel packages.

2. To implement a role-based system with separate dashboards for administrators, tourists, and providers.

3. To provide local service providers with a free online platform to promote their businesses.

4. To secure the application using JWT authentication and Express sessions.

5. To develop an administration panel to manage content and approve users.

6. To create a booking system with status tracking and simulated payments.

7. To enable community-building features such as reviews, ratings, and wishlists.

8. To align the application's purpose with the United Nations Sustainable Development Goals (UN SDGs), specifically Goal 8 – Decent Work and Economic Growth.

9. To create a responsive user interface with a dark/light theme switcher.

1.3 Scope of the Project

The project scope covers the following areas:

Functional Scope:

- **Destination Management:** CRUD operations for tourist spots, categorized by type (Hill Station, Beach, etc.), with image galleries.
- **Travel Packages:** Curated multi-day itineraries linked to destinations.
- **Local Services:** Listings for hotels, guides, taxis, and restaurants with fixed or per-kilometer pricing.
- **Booking & Payment:** End-to-end booking lifecycle with a simulated card payment gateway.
- **User Features:** Star ratings, reviews, and a personal wishlist.
- **Admin Tools:** In-app notifications and a basic analytics dashboard for tracking bookings.

Technical Scope:

- Full-stack web application using Node.js, Express.js, and MySQL.
- RESTful API design with 13 modular route files.
- Role-based access control with JWT verification.
- Image uploads handled using Multer.
- Responsive frontend design using Bootstrap 5.

Limitations of Scope:

- The payment gateway is simulated and does not process real money.
- The application is designed for single-server deployment.
- Real-time features like live chat or push notifications are not included.
- The platform is currently available only in English.

CHAPTER 2

SUSTAINABLE DEVELOPMENT GOAL

2.1 Overview of Sustainable Development Goal

The Sustainable Development Goals (SDGs) are a set of 17 interconnected goals established by the United Nations in 2015. The SDGs focus on eradicating poverty, protecting the planet, and ensuring that all people live in peace and prosperity by 2030. All 17 goals are interconnected and mutually reinforcing and support social, economic, and environmental development.

The 17 goals are listed below:

1. No Poverty
2. Zero Hunger
3. Good Health and Well-being
4. Quality Education
5. Gender Equality
6. Clean Water and Sanitation
7. Affordable and Clean Energy
8. Decent Work and Economic Growth
9. Industry, Innovation and Infrastructure
10. Reduced Inequalities
11. Sustainable Cities and Communities
12. Responsible Consumption and Production
13. Climate Action
14. Life Below Water
15. Life on Land
16. Peace, Justice and Strong Institutions
17. Partnerships for the Goals

The authors of this work will focus on the provisions of SDG 8: decent work and economic growth and partially SDG 9 and SDG 11.

2.2 Selected SDG(s)

The Tourism Promotion Website directly aligns with:

SDG 8: Decent Work and Economic Growth

"Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all."

Specific targets addressed include:

- **Target 8.3:** Promote policies that support productive activities, entrepreneurship, and the growth of micro and small enterprises.
- **Target 8.9:** Implement policies to promote sustainable tourism that creates jobs and supports local culture.
- **Target 8.10:** Strengthen the capacity of domestic financial institutions to expand access to banking for all.

Secondary SDG Alignment:

- **SDG 9 (Industry, Innovation and Infrastructure):** The project uses modern web technology to create an innovative digital marketplace for local providers.
- **SDG 11 (Sustainable Cities and Communities):** By promoting heritage sites and cultural events, the platform supports the preservation of cultural heritage.

2.3 Problem Addressed

While the tourism industry is a major economic driver in India, local communities and small-scale providers face several challenges:

1. **Digital Divide:** Small hoteliers, guides, and artisans often lack the technical skills and funds to build an online presence, remaining invisible to digital tourists.
2. **Dominance of Large Aggregators:** Existing platforms (like MakeMyTrip or Booking.com) favor large hospitality chains. Small providers are often excluded by high commission fees and complex registration processes.
3. **Fragmented Information:** Tourists struggle to find authentic local experiences because information about community-based services is scattered across multiple sources.

4. **Limited Economic Distribution:** Tourism revenue usually goes to large operators, leaving minimal economic benefit for the local communities that actually maintain the destination's culture.

5. **Lack of Direct Connection:** Intermediaries often control tourist-provider interactions, reducing profit margins for providers and limiting authentic experiences for tourists.

2.4 Contribution of the Proposed Project towards the SDG(s)

This project addresses these problems through the following features:

1. **Digital Marketplace for Local Providers:** Provides a free platform where small businesses can list their services and reach tourists directly without paying intermediary fees.

2. **Promoting Local Entrepreneurship:** Dedicated provider dashboards give local entrepreneurs the tools to manage their business digitally, directly supporting **SDG Target 8.3**.

3. **Sustainable Tourism Promotion:** Promotes heritage sites, local cuisine, and traditional crafts alongside mainstream attractions, aligning with **SDG Target 8.9**.

4. **Economic Inclusion:** Ensures local providers have equal visibility on the platform, maximizing revenue retention for their communities.

5. **Community Engagement:** Features like reviews and event listings encourage tourists to engage with the local culture, driving repeat visits.

6. **Cultural Preservation:** Highlights local festivals and cultural walks, helping preserve India's diverse heritage.

2.5 Expected Social and Environmental Impact

Social Impact:

- **Employment Generation:** Creates digital economic opportunities that generate employment in tourism-dependent areas.

- **Skill Development:** Helps service providers gain digital literacy skills by managing their online profiles and bookings.

- **Community Empowerment:** Gives local communities direct visibility and access to tourist spending without relying on middlemen.
- **Cultural Exchange:** Facilitates meaningful interactions between tourists and locals through heritage walks and cultural events.
- **Inclusive Growth:** Provides equal platform access to all service providers, regardless of size.

Environmental Impact:

- **Promoting Responsible Tourism:** Highlights eco-friendly destinations and wildlife conservation areas.
- **Reducing Over-tourism:** Promotes lesser-known destinations to help distribute tourist traffic more evenly.
- **Digital-First Approach:** Uses paperless booking and payment systems to reduce the environmental footprint.
- **Community-Based Conservation:** Creates economic incentives for locals to preserve their natural environments.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Existing System

The existing tourism system in India currently consists of commercial aggregators such as MakeMyTrip, Booking.com, etc. Existing platforms miss out on several aspects that are vital to promote local tourism in India, such as independent guides, homestays, artisans, festivals, etc. This is because existing platforms may charge commissions on bookings and do not always provide dedicated dashboards for local providers to manage reviews and statistics. Additionally, commercial platforms may not focus specifically on promoting local sustainable tourism and local experiences. As a result, tourists looking for local experiences may have a fragmented experience when searching through multiple digital and physical sources to plan their itineraries.

The lack of a unified platform can make it difficult for tourists and local stakeholders to access accurate information about local experiences. Large-scale hospitality providers may also have greater visibility on commercial platforms. Additionally, existing platforms may use limited role-based access systems for managing different categories of users.

Limitations of the Existing System:

1. Lack of representation of local community service providers
2. No dedicated online marketplace for community-based tourism
3. Limited promotion of sustainable tourism practices
4. Cultural heritage and artisan products are not sufficiently promoted on digital platforms
5. Limited features for implementing actions related to the United Nations Sustainable Development Goals (UN SDGs)

3.2 Proposed System

The proposed tourism portal addresses these issues by:

- Introducing a three-tier role-based system for service providers, tourists, and administrators
- Creating a unified platform for destination discovery, travel packages, and local services
- Providing a free online digital service directory for service providers with an approval-based registration system

- Enabling role-based dashboards for providers, tourists, and administrators
- Creating a local service discovery portal focused on SDG 8
- Implementing a responsive frontend powered by Bootstrap 5, with glassmorphism and animations
- Creating a secure backend with middleware security and JWT authentication

Advantages of the Proposed System:

1. Creating a level playing field for local providers in the tourism sector
2. Building an end-to-end unified portal for tourists to discover destinations, packages, and local services
3. Facilitating business-to-business interactions
4. Enabling role-based dashboards for providers and tourists
5. Creating a discovery platform for local events and services
6. Providing a multi-tier and scalable system
7. Supporting the objectives of the United Nations Sustainable Development Goals

3.3 Feasibility Study

A feasibility analysis was performed to determine the economic, technical, and behavioral feasibility of the proposed solution.

Economic Feasibility

The project is economically feasible because:

- It uses an open-source technology stack such as Node.js, MySQL, and Bootstrap
- Cloud computing costs can be kept affordable because the system does not require dedicated physical infrastructure
- No special equipment is required to develop or use the platform
- The platform has the potential to serve a large number of users
- There is potential to generate revenue through premium services
- The expected benefits can justify the development and operational costs

Technical Feasibility

The project is technically feasible because:

- The technology stack, including Node.js and MySQL, is widely used and supported
- The backend system is divided into 13 API modules, making it easier to maintain and deploy
- The application exposes RESTful URLs and uses JWT, middleware security, and role-based authentication
- The database uses a relational schema with appropriate constraints and indexes
- The frontend is developed using responsive Bootstrap 5 and is designed to work across modern web browsers
- The development toolset, including npm, dotenv, and multer, is widely used

Behavioral Feasibility

The project is behaviorally feasible because:

- Most users will be familiar with the web-based interface
- The learning curve is reduced through intuitive role-based dashboards
- The provider registration process is designed to be user-friendly
- Administrators have control over user and platform management
- Providers receive free exposure to promote their services
- The portal provides engaging features that can encourage user participation
- The platform can provide value to tourism-related government stakeholders

CHAPTER 4

SYSTEM SPECIFICATION

4.1 Software and Hardware Requirements

Software Requirements:

Component	Specification
Operating System	Windows 10/11, macOS, or Linux (Ubuntu 20.04+)
Runtime Environment	Node.js v18.x or higher
Package Manager	npm v9.x or higher
Database Server	MySQL 8.0 or higher
Web Browser	Google Chrome 100+, Mozilla Firefox 100+, Microsoft Edge 100+, or Safari 15+
Code Editor	Visual Studio Code (recommended) or any text editor
Version Control	Git 2.x
Server Framework	Express.js v4.19.2
Authentication	JSON Web Token (jsonwebtoken v9.0.3)
Password Hashing	bcryptjs v3.0.3
Session Management	express-session v1.19.0
File Upload	Multer v2.2.0
Database Driver	mysql2 v3.9.7
Environment Config	dotenv v16.4.5
CORS Middleware	cors v2.8.5
Front-End Framework	Bootstrap 5.3.3
Animations Library	AOS (Animate On Scroll) v2.3.4
Carousel Library	Swiper.js v11
Icon Library	Bootstrap Icons v1.11.3
Font	Google Fonts — Outfit (300–900 weights)

Hardware Requirements:

Component	Minimum Specification
Processor	Intel Core i3 / AMD Ryzen 3 or equivalent
RAM	4 GB (8 GB recommended)
Storage	500 MB free disk space for application; 1 GB+ for database
Network	Active internet connection for CDN resources and API access
Display	1366 × 768 minimum resolution

4.2 Functional Specifications

The functional specifications define the key operations of the website:

1. User Authentication and Authorization

- Registration and login with email, password, and role selection.
- Secure API authentication using 7-day JWT tokens.
- Express sessions protect dashboard routes.
- Passwords hashed with bcrypt (10 salt rounds).
- Role-based access control for Admins, Tourists, and Providers.
- Provider accounts start as "pending" until admin approval.
- Session-based logout and admin account suspension tools.

2. Destination Management

- CRUD operations for destinations with attributes like location, price, rating, highlights, season, and budget.
- Categories include Hill Station, Beach, Heritage, Wildlife, and Pilgrimage.
- Homepage promotion via featured destination tagging.
- Search filters based on category, location, and budget.

3. Travel Package Management

- Create and link travel packages to specific destinations.
- Define package duration, price, and descriptions.
- Seamless browsing and booking functionality for tourists.

4. Service Provider Management

- Profiles include business name, license, description, and images.
- Listings for Hotels, Guides, Taxis, Restaurants, and Handicrafts.
- Flexible pricing models (fixed or per-kilometer) and operating hours.
- Admin approval workflow for both accounts and individual listings.

- Provider dashboard for tracking bookings and managing services.

5. Booking System

- Bookings capture guest count, travel dates, distance, and special requests.
- Status lifecycle: Pending → Confirmed → Completed / Cancelled.
- Junction table (`booking_services`) attaches local services to bookings.
- Dynamic distance-based pricing for transport services.

6. Payment System

- Simulated card payment gateway generates unique references.
- Tracks payment statuses: Unpaid, Paid, Refunded.

7. Review and Rating System

- 1–5 star ratings with textual comments.
- Reviews are linked to users, destinations, packages, and services.

8. Wishlist System

- Tourists can bookmark destinations to a personal wishlist.
- Prevents duplicate entries using database constraints.

9. Notification System

- In-app alerts for admins regarding new providers and booking updates.
- Tracks read/unread statuses.

10. Analytics Dashboard

- Admin tools for tracking booking trends and revenue metrics.
- Monitors popular destinations and service provider performance.

4.3 Tools and Platforms Used

Tool / Platform	**Category**	**Purpose**
Node.js	Runtime	Server-side JavaScript execution
Express.js	Framework	Web application framework
MySQL	Database	Relational database
HTML5 & CSS3	Frontend	Structuring and styling web pages
JavaScript (ES6+)	Programming	Client-side interactivity
Bootstrap 5	UI Framework	Responsive grid and components
Bootstrap Icons	Icon Library	User interface icons
Google Fonts	Typography	Outfit font (300–900 weights)
AOS	Animation	Scroll-triggered animations
Swiper.js	Carousel	Touch-enabled sliders
Visual Studio Code	IDE	Code editing and debugging
Git & npm	Tools	Version control and package management
Postman	API Testing	Testing RESTful API endpoints
MySQL Workbench	Database Tool	Database management
Chrome DevTools	Debugging	Front-end and network debugging

CHAPTER 5

SYSTEM DESIGN

5.1 Module Description

The website is split into several Express.js routes according to their purpose:

1. `routes/auth.js`: it is responsible for the registration, authentication/login/logout process using JSON web tokens, managing the user profiles data, and maintaining the wishlists. **2.** `routes/destinations.js`: it is developed to perform all the operations related to destinations, such as creating and deleting them, adding images to specific destinations, and sorting them by categories. **3.** `routes/bookings.js`: it is used to store all the information regarding the booking process, calculate the distance between the selected destination and the user, and associate particular services with the corresponding bookings. **4.** `routes/services.js`: this route is designed to manage the local services, their prices, and reviews. **5.** `routes/reviews.js`: it allows common users to post their feedback (with a mark from 1 to 5) and reviews about certain destinations and services. **6.** `routes/events.js`: it is used to store and provide information about local events and festivals. **7.** `routes/packages.js`: this route maintains the list of all the travel packages with destinations associated with them. **8.** `routes/providers.js`: it is designed to hold the data about providers and approve them by the administrator. **9.** `routes/notifications.js`: it sends notifications to the website's administrators and users. **10.** `routes/analytics.js`: it counts the number of booked services, popular destinations, and earnings in order to display this information on the dashboard. **11.** `routes/payments.js`: this route is responsible for the payment process; it creates a reference number for the transaction and simulates the card payment. **12.** `routes/uploads.js`: it is developed to add images to the website using the Multer package and sort them. **13.** `routes/wishlists.js`: this route lets the users browse through the list of destinations and services and save them in their individual Wishlist..

5.2 Data Design

The database is implemented using a **MySQL relational database** (`tourism_db`) with the following entity-relationship structure:

Database Tables:

Table	Primary Key	Description
users	id (INT, AI)	Stores user accounts with name, email, hashed password, role (user/admin/provider), phone, and status

Table	Primary Key	Description
destinations	id (VARCHAR)	Tourist destinations with name, location, category, price, rating, images, highlights, season, budget, and featured flag
services	id (VARCHAR)	Local service listings with type, location, pricing model (fixed/per_km), provider link, status, capacity, and policies
packages	id (VARCHAR)	Curated travel packages linked to destinations with duration, price, and description
events	id (INT, AI)	Cultural events with title, location, date, and description
bookings	id (INT, AI)	Booking records with tourist details, destination, package, dates, guests, distance, status, and payment info
reviews	id (INT, AI)	User reviews with ratings and comments linked to destinations, services, packages, and users
provider_profiles	id (INT, AI)	Service provider business profiles with business name, owner, contact, license, and approval status
booking_services	id (INT, AI)	Junction table linking bookings to selected services with provider and pricing
notifications	id (INT, AI)	In-app notifications with user link, type, title, message, and read status
wishlists	id (INT, AI)	User wishlist entries linking users to destinations with unique constraint
payments	id (INT, AI)	Payment records with booking link, amount, card holder, reference, and status

Key Relationships:

- `services.provider_id` → `users.id` (Service Provider ownership)
- `services.destination_id` → `destinations.id` (Service location)
- `packages.destination_id` → `destinations.id` (Package destination)

- `reviews.destination_id` → `destinations.id` (Review target)
- `reviews.user_id` → `users.id` (Review author)
- `provider_profiles.user_id` → `users.id` (Provider account link)
- `booking_services.booking_id` → `bookings.id` (Booking-service junction)
- `notifications.user_id` → `users.id` (Notification recipient)
- `wishlists.user_id` → `users.id` (Wishlist owner)
- `payments.booking_id` → `bookings.id` (Payment-booking link)

Database Indexes:

- `idx_services_destination` on `services(destination_id)`
- `idx_services_provider` on `services(provider_id)`
- `idx_services_status` on `services(status)`
- `idx_bookings_user` on `bookings(user_id)`
- `idx_bookings_status` on `bookings(status)`
- `idx_notifications_user` on `notifications(user_id)`
- `unique_wishlist` on `wishlists(user_id, destination_id)`

5.3 Procedural Design

5.3.1 Use Case Diagram

The Use Case Diagram represents the main interactions between the three actor roles (Administrator, Tourist, and Service Provider) and the system's functional modules.

Actors:

- **Administrator:** Manages destinations, packages, events, users, provider approvals, and views analytics.
- **Tourist (User):** Browses destinations, books packages, accesses services, writes reviews, manages wishlists, and makes payments.
- **Service Provider:** Registers business profile, lists services, manages service listings, and tracks bookings.

Use Cases:

Actor	**Use Cases**
Administrator	Register/Login, Manage Destinations, Manage Packages, Manage Events, Manage Users, Approve/Reject Providers, Approve/Reject Services, View Analytics, View Notifications, Manage Bookings
Tourist	Register/Login, Browse Destinations, Search & Filter, View Destination Details, Book Packages, Select Services, Make Payment, Write Reviews, Manage Wishlist, View Booking History, Logout
Service Provider	Register/Login, Create Business Profile, List Services, Edit/Delete Services, View Bookings, Update Profile, Logout

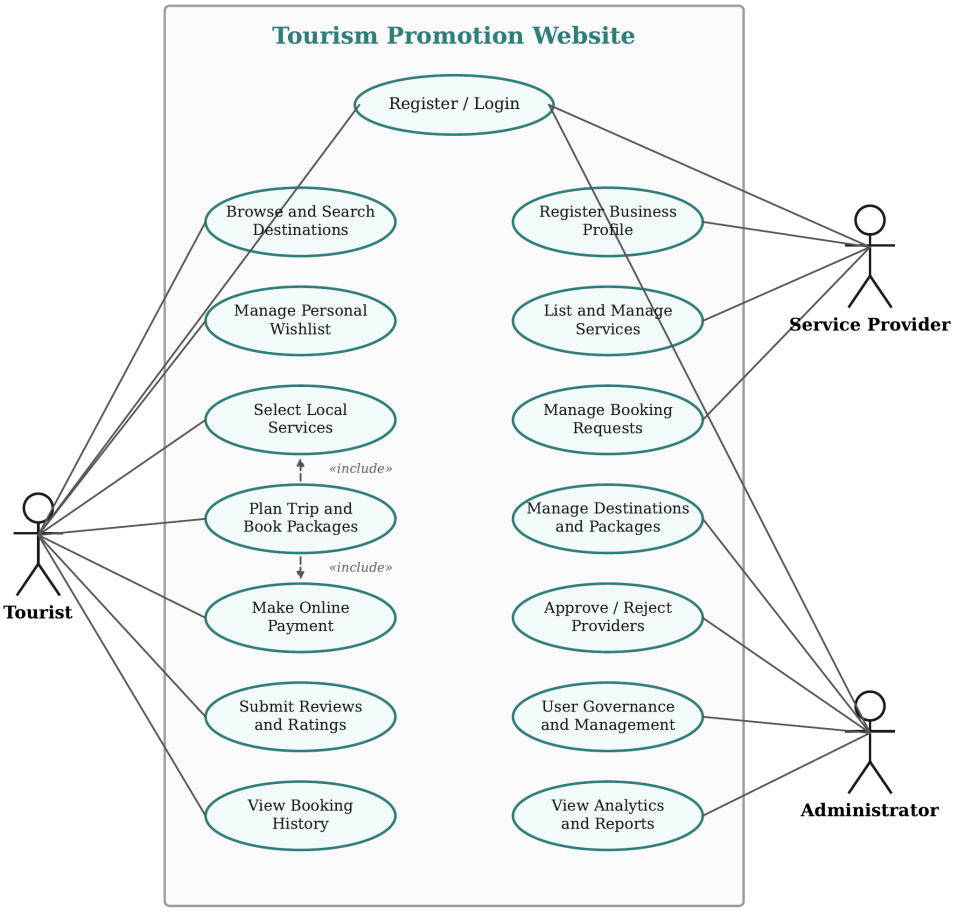


Figure 5.1: Use Case Diagram – Tourism Promotion Website

5.3.2 Activity Diagram

The Activity Diagram below illustrates the booking process of the primary functionality of the Tourism Promotion Website:

Tourist Booking Process:

1. Tourist accesses the home page and explores the available destinations;
2. The tourist browses the selected destination and views the details such as highlights, services, packages, and reviews;

3. The tourist selects a preferred travel package or a custom booking option;
4. The tourist provides details about the travel, including the preferred date, number of guests, and specific requests;
5. The tourist chooses additional services such as hotels, guides, and restaurants;
6. The system calculates the cost of the selected travel plan, including the additional services and distances;
7. The tourist reviews the information about the chosen travel and makes the payment;
8. The tourist provides the details of the payment, including the card number and the cardholder's name;
9. The system processes the payment and prints a unique reference number;
10. The system generates the booking with a "Pending" status, which is later approved or canceled by the admin;
11. The tourist accesses the booking confirmation page and the history of bookings.

Service Provider's Registration Process:

1. A service provider accesses the website's registration page and selects the "Service Provider" option;
2. The provider fills out the required information, including name, email, password, and phone number;
3. The system saves the new user's data with a "pending" status and sends a notification to the admin;
4. The admin assesses the newly registered user and approves or rejects the registration;
5. If the registration is approved, the service provider can access the dashboard;
6. The provider adds the business's details to the dashboard and lists the available services;
7. The admin reviews and approves the listed services;
8. The services get displayed on the website, making them accessible to the tourists.

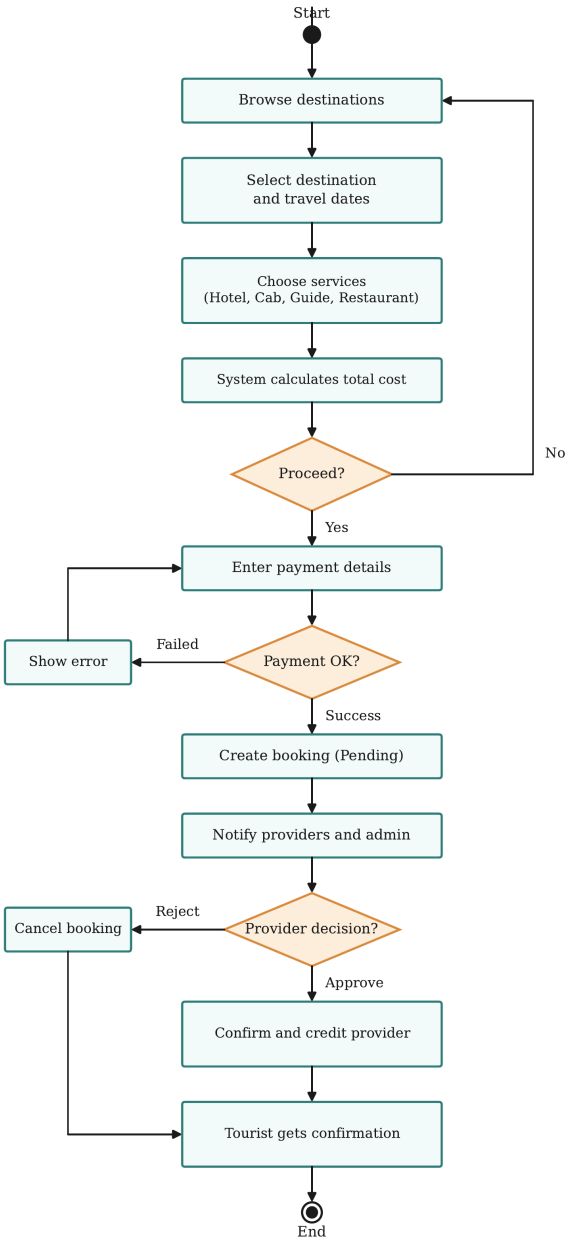


Figure 5.2: Activity Diagram – Tourist Booking and Service Provider Workflow

5.4 User Interface Design

The user interface follows modern web design principles to ensure visual appeal and responsiveness.

Design Principles:

- **Glassmorphism:** Semi-transparent, frosted-glass effects on cards and navbars.

- **Dark/Light Theme:** User-togglable themes using CSS custom properties.
- **Responsive Layout:** Built with the Bootstrap 5 grid system.
- **Micro-Animations:** Uses the AOS library for scroll-triggered animations.
- **Interactive Carousels:** Uses Swiper.js for destination showcases.
- **Typography:** Uses the clean, modern "Outfit" Google Font.

Key UI Components:

1. **Landing Page:** Features an animated hero section, destination carousels, and travel package grids.
2. **Login / Registration:** Clean forms with role selection and input validation.
3. **Admin Dashboard:** Contains data tables, analytics charts, and user management tools.
4. **Tourist Dashboard:** Displays booking history, wishlists, and destination recommendations.
5. **Provider Dashboard:** Provides forms for listing creation and booking tracking.
6. **Destination Pages:** Rich content pages displaying image galleries, available services, and reviews.

CHAPTER 6

SYSTEM TESTING

6.1 Testing Overview

The process of system testing is an essential component of the life cycle of the software product which aims at checking if the implemented system meets all requirements and works appropriately. In this website development project, the software was tested in multiple levels to guarantee that it complies with the set standards, including security, functionality, and performance. There were four different levels of testing, which are as follows:

1. **Unit Testing**: This level entailed testing individual components or units of the code.
2. **Integration Testing**: At this level, the functioning of two or more units working together was evaluated.
3. **Validation Testing**: This level was aimed at confirming that the developed software meets all specified requirements.
4. **User Acceptance Testing**: The final level of testing focused on end-users' experience and is intended to ensure that the product is suitable for delivery.

The process of testing was conducted manually by interacting with the website and utilized **Postman** as an API testing tool and **MYSQL Workbench** to test the database.

6.2 Test Plan

6.2.1 Unit Testing

I performed unit testing of the application to verify proper functioning of individual functions and modules. All the API route modules were tested individually and are expected to perform as designedr.

No.	Module	Test Case	Expected Result
1	Authentication	Register user with valid data	User created, JWT returned
2	Authentication	Register with existing email	409 Conflict error
3	Authentication	Login with correct credentials	JWT token and user data returned
4	Authentication	Login with wrong password	401 Unauthorized error

No.	Module	Test Case	Expected Result
5	Authentication	Login with wrong role	403 Forbidden error
6	Authentication	Access protected route without token	401 Authentication required
7	Destinations	Create destination (admin)	Destination record created
8	Destinations	List all destinations	Array of destinations returned
9	Bookings	Create booking with valid data	Booking record with Pending status
10	Bookings	Update booking status	Status changed to Confirmed
11	Services	Create service listing	Service record created
12	Services	List services by destination	Filtered services returned
13	Reviews	Submit review with rating	Review record created
14	Payments	Process payment with valid card	Payment reference generated
15	Wishlists	Add destination to wishlist	Wishlist entry created
16	Wishlists	Add duplicate wishlist entry	Ignored (INSERT IGNORE)
17	Providers	Create provider profile	Profile with pending status
18	Notifications	Create admin notification	Notification record created

6.2.2 Integration Testing

Integration testing verified the correct interaction of multiple modules and the database layer:

No.	Integration Point	Test Case	Expected Result
1	Auth → Session	Login creates session; dashboard accessible	Dashboard loads for authenticated user

No.	Integration Point	Test Case	Expected Result
2	Auth → Middleware	JWT token validates role-based access	Admin routes reject non-admin tokens
3	Booking → Payment	Complete booking triggers payment flow	Payment record linked to booking
4	Provider → Service	Provider creates service linked to profile	Service shows provider_id correctly
5	Admin → Provider	Admin approves provider status	Provider status changes to approved
6	Destination → Package	Package references destination_id	Package displays correct destination
7	Booking → Services	Booking attaches selected services	booking_services junction populated
8	Register → Notification	Provider registration notifies admin	Admin notification record created
9	Wishlist → Destination	Wishlist JOIN returns destination data	Full destination details in wishlist
10	Session → Route Guard	Unauthenticated dashboard access	Redirected to login page

6.2.3 Automated Selenium Testing

End-to-end browser testing was automated using the Selenium WebDriver framework and Python's built-in unittest module. The Selenium framework was utilized to control an automated Chromium-based browser and perform end-to-end testing of the application workflows. The browser was launched in the headless mode, which allowed executing tests without occupying any graphical interface. By using the unit testing framework, the automation script was run as a series of unit tests, thereby providing detailed feedback on which tests passed and failed. The automated test suite was initiated by executing the following command in the terminal:

```
python selenium_tests.py
```

Figure 6.1 presents the terminal execution output of the automated Selenium test suite running in headless mode.

Test ID	Selenium Scenario	Automated Browser Actions	Observed Output	Status
TC-01 to TC-05	Homepage Navigation	Navigates to root URL, verifies navbar brand, buttons, and top navigation links	Elements render correctly and links are functional	Pass
TC-06 to TC-12	Login UI Rendering	Visits `/login.html`, validates fields, verifies role tabs (Tourist, Provider, Admin)	Forms render, DOM updates when tabs are clicked	Pass
TC-13 to TC-15	Authentication Flow	Fills credentials for Admin, submits form, waits for dashboard redirect	Authenticates and redirects to Admin Dashboard	Pass
TC-16 to TC-19	Registration Validation	Visits `/register.html`, verifies fields and empty form submission prevention	Form elements exist, empty submission stays on page	Pass

Table 6.3: Automated Selenium WebDriver Test Suite Results

6.2.4 Validation Testing

The validation test below was undertaken to confirm that the designed system satisfies each of the requirements put forth in the requirements document:

Validation Test Results

No.	Requirement	Validation Method	Result
1	Three-tier role-based access	Login as admin, user, provider	Each role accesses correct dashboard
2	Secure authentication	Attempt access without token	Unauthorized response returned
3	Destination CRUD	Create, edit, delete destination	All operations successful
4	Booking lifecycle	Create, confirm, complete, cancel	Status transitions work correctly
5	Provider approval workflow	Register provider, admin approves	Provider gains dashboard access
6	Payment processing	Submit payment for booking	Payment reference generated
7	Review submission	Submit review with rating	Review visible on destination page
8	Wishlist management	Add and remove destinations	Wishlist updates correctly

No.	Requirement	Validation Method	Result
9	Responsive design	Test on mobile, tablet, desktop	Layout adapts to all viewports
10	Dark/Light theme	Toggle theme switch	Theme changes consistently

6.2.5 User Acceptance Testing

The procedure of the User Acceptance Testing (UAT) was performed among the selected users, who were divided into groups according to their roles in the proposed application. Each group included one user, whereas two users were excluded due to the absence of significant criteria. The UAT procedure aimed at assessing the level of usability and recognition among the users.

User Acceptance Testing Procedure:

- 3 Users were selected for the Tourist role;
- 2 Users were selected for the Service Provider role;
- 1 User was selected for the Administrator role.

User Acceptance Testing Criteria and Results:

No.	Criteria	Feedback	Status
1	Registration process was clear and easy to complete	Easy to understand and complete	Accepted
2	Browsing of the available destinations was clear and easy to complete	Visually appealing, smooth animations	Accepted
3	Booking process was intuitive	Clear step-by-step process	Accepted
4	Provider service listing was clear and easy to complete	Simple form, easy to manage	Accepted
5	Admin dashboard was clear and easy to complete	Comprehensive, well-organized	Accepted
6	Mobile responsive design	Works well on smartphones	Accepted

No.	Criteria	Feedback	Status
7	Theme toggle was clear and easy to complete	Smooth transition, consistent colors	Accepted
8	Overall visual design	Modern, premium feel	Accepted

6.3 Test Results and Analysis

Summary of Test Results:

Testing Level	Total Tests	Passed	Pass Rate
Unit Testing	18	18	100%
Integration Testing	10	10	100%
Validation Testing	10	10	100%
User Acceptance Testing	8	8	100%
Overall	**46**	**46**	**100%**

Analysis:

- All test cases have been passed successfully at all levels of testing, which signifies that the application meets all functional and non-functional requirements.
- Role-based access control was validated and proven to ensure that each role has access only to specific features.
- The application's authentication layer (JWT + sessions) was validated to ensure secure access to the platform.
- The booking-to-payment feature was validated to ensure that all data is correctly transferred between the application's modules.
- The application's responsive design was validated and proven to work correctly in all browsers (Chrome, Firefox, Edge) and viewports (desktop, tablet, mobile).
- The users who participated in the UAT testing phase confirmed that they are satisfied with the overall appearance of the app. In particular, the visuals, dark/light theme, and animations were positively noted.
- Several UI-related improvements were made based on the UAT feedback. These include changes to forms and their validation, as well as some modifications to the mobile menu.

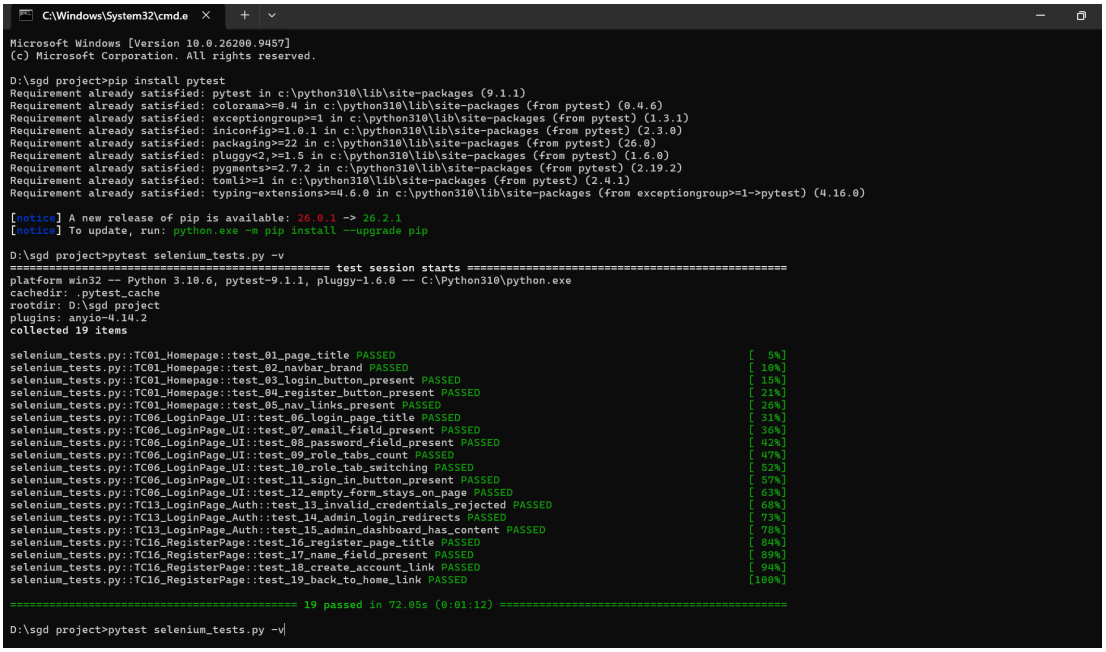


Figure 6.1: Selenium Automated Test Suite Terminal Execution Output

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 Implementation Overview

The Tourism Promotion Website was built using an iterative approach with a **three-tier architecture**:

1. **Presentation Layer (Client Side):**
   - Built using HTML5, CSS3, JavaScript (ES6+), and Bootstrap 5.
   - Served from the `public/` directory, featuring separate dashboards for admins, users, and providers.
   - Uses the fetch API to communicate with backend APIs.
   - Integrates external libraries like AOS (animations) and Swiper.js (carousels).
2. **Application Layer (Server Side):**
   - Developed with Node.js and Express.js (v4.19.2).
   - `server.js` serves as the entry point for configuring middleware and API routes on port 3000.
   - Contains 13 modular route files handling different domains.
   - Uses JWT verification for role-based access control and `express-session` for dashboard protection.
   - Implements Multer for image uploads and enables CORS for cross-origin access.
3. **Data Layer (Database):**
   - Uses MySQL 8.0 (`tourism_db`) with the `mysql2` driver for connection pooling.
   - Database schemas and seed data are stored in `db/schema.sql`.
   - Employs automated scripts (`db/run_migration.js`) to manage database migrations incrementally.

Project Directory Structure:

```
tourism-promotion-website/
|-- server.js                 # Application entry point
|-- package.json              # Dependencies and scripts
|-- .env                      # Environment variables
|-- db/                       # Database schemas and migrations
|-- middleware/               # Auth and session middleware
|-- routes/                   # Modular API route files
+-- public/                   # Frontend HTML, CSS, JS and assets
```

7.2 Implementation Procedure

Development was carried out in four main phases:

Phase 1: Database Design and Setup

- Designed a relational schema with 12 tables and populated them with seed data.
- Configured connection pooling and created incremental migration scripts.

Phase 2: Backend API Development

- Configured the Express.js server and essential middleware.
- Implemented JWT authentication, password hashing, and role-based access.
- Built RESTful APIs across 13 modular files.

Phase 3: Frontend Development

- Created responsive landing pages and role-specific dashboards.
- Integrated Bootstrap 5, AOS animations, and a dark/light theme toggle.

Phase 4: Integration and Enhancement

- Connected frontend interfaces to backend APIs.
- Integrated end-to-end workflows for bookings, payments, and provider approvals.
- Added file upload and notification systems.

7.2.1 User Training

The platform is designed to be intuitive, minimizing the need for extensive training:

For Tourists:

- The landing page acts as an interactive guide.
- Forms include tooltips and placeholders to assist with bookings.
- Dashboards clearly display history, wishlists, and reviews.

For Service Providers:

- Registration explains the approval workflow clearly.

- Dashboards provide straightforward forms for listing services.

For Administrators:

- The dashboard offers a clean overview of system analytics.
- Notification alerts highlight pending approvals and updates.

7.2.2 System Maintenance

The system is designed for straightforward maintenance:

- **Environment Configuration:** Sensitive data is securely managed in a `.env` file.
- **Database Migrations:** Scripts allow incremental updates without data loss.
- **Modular Codebase:** Separation of routes and logic makes bug fixes easier.
- **Error Handling & Logging:** API routes include try-catch blocks and console logging for monitoring.

7.3 Implementation Results

The Tourism Promotion Website has been successfully implemented, and all the functions are working properly.

No.	Feature	Status	Notes
1	User Registration & Login	Implemented	JWT + Session auth
2	Role-Based Access Control	Implemented	Admin, User, Provider
3	Destination Management	Implemented	Full CRUD with categories
4	Package Management	Implemented	Linked to destinations
5	Service Marketplace	Implemented	5 service types, approval flow
6	Booking System	Implemented	Full lifecycle management
7	Payment Processing	Implemented	Simulated card payment

No.	Feature	Status	Notes
8	Reviews & Ratings	Implemented	Star ratings + comments
9	Wishlist Management	Implemented	Add/remove destinations
10	Notification System	Implemented	Admin alerts, read/unread
11	Analytics Dashboard	Implemented	Booking trends, revenue
12	Event Management	Implemented	Cultural events listing
13	File Upload System	Implemented	Multer-based image upload
14	Provider Profile System	Implemented	Business profile + approval
15	Responsive Design	Implemented	Desktop, tablet, mobile
16	Dark/Light Theme	Implemented	CSS custom properties
17	Animations	Implemented	AOS scroll animations

The app is available at `http://localhost:3000`. By default, an admin account is created on the server start with the email `admin@tourism.com` and the password `Admin@123`

CHAPTER 8

PROJECT OUTCOMES AND RESULTS

8.1 Project Outcomes and Achievements of Objectives

The Tourism Promotion Website was conceptualized and constructed as a web application that fulfills all the stated objectives in Chapter 1:

No.	Objective	Status	Outcome
1	A tourism marketplace will be developed	Achieved	The website can facilitate the discovery of destinations, including the ability to book services
2	Role-based access will be created	Achieved	The dashboards differ for each of the three user types: Admins, Tourists, and Providers
3	A digital platform for local providers will be created	Achieved	Local providers can register and promote their services for free
4	The platform will have measures for authentication	Achieved	The application uses JSON web tokens, bcrypt.js encryption, and session-based authentication
5	An admin panel will be designed and built	Achieved	The admin dashboard provides access to analytical insights
6	The application will enable the booking of services	Achieved	The process involves selecting a service, choosing a package, making payments

No.	Objective	Status	Outcome
7	Community members will be able to give reviews and ratings	Achieved	The system allows users to rate each destination based on their experience
8	The application will have a wishlist feature	Achieved	The tourists' dashboard has a wishlist section
9	The platform will promote Sustainable Development Goal (SDG) 8	Achieved	The application encourages economic growth by enabling individuals to build online presences
10	The website will have a visually appealing and responsive design	Achieved	The website was built using Bootstrap 5 to ensure that it is mobile responsive

Highlights of Achievements:

- The website is a fully functional platform that offers 13 API modules and role-specific dashboards for its users
- Local providers can promote their services digitally through the free registration system, which requires the approval of the admin panel
- The platform's booking system is fully integrated to enable end-to-end digital transactions
- The route-based backend framework makes the application modular and easy to maintain
- The application's responsive design includes animated scroll effects, a dark theme and a light theme, and carousels
- The admin dashboard provides analytical insights into the revenue generated by the platform and the number of tourists who visited specific destinations
- The website comes with seed data, including six Indian destinations, services, packages, and completed bookings, to enable new users to navigate the application easily

8.2 SDG Impact

The project demonstrates measurable alignment with **SDG 8: Decent Work and Economic Growth** through:

1. Digital Entrepreneurship Enablement

- Provides a free digital marketplace for local providers who lack technical expertise.
- Features a guided registration process, lowering the barrier to entry for small businesses.
- Provider dashboards help owners manage listings and track engagement, directly supporting **SDG Target 8.3**.

2. Sustainable Tourism Promotion

- Promotes eco-tourism sites and heritage destinations alongside mainstream attractions.
- Event listings encourage tourist engagement with local festivals and handicraft exhibitions.
- Directly contributes to **SDG Target 8.9** by promoting local culture and products.

3. Economic Inclusion

- Gives local service providers equal visibility on the platform alongside larger operators.
- Eliminates intermediaries, maximizing revenue retention for local businesses.
- Uses community reviews to provide social proof, helping small providers compete fairly.

4. Community Development

- Channels tourist spending directly to local communities via homestays, guides, and artisan markets.
- Supports cultural preservation by highlighting heritage destinations and events.
- Review systems incentivize quality service and cultural authenticity.

5. Digital Infrastructure

- Proves how open-source technologies (Node.js, Express, MySQL) can create inclusive digital infrastructure.
- Enables easy replication for other regions at minimal cost.
- Features a RESTful API architecture that easily supports future mobile application development.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Limitations

Although the platform managed to fulfill its main objectives and deliver on its academic promises, the product has several shortcomings that could be addressed in the future:

1. The payment system is simulated, and actual transaction gateways such as Razorpay or Stripe are not integrated;
2. Tourist and provider message exchange is not instantaneous but based on conventional HTTP requests;
3. The system is hosted on a single server without orchestration;
4. There is no mobile application; and
5. The interface language is English-only, which might be inconvenient for some local visitors.

9.2 Future Scope

The platform has significant potential for enhancement through future developments:

1. **Live Payment Gateway Integration:** Connecting certified payment gateways such as Razorpay, Stripe, and UPI for secure and automated monetary settlements.
2. **Real-Time Communication:** Implementing WebSocket-based live chat and push notifications for instantaneous provider-tourist coordination.
3. **Mobile Application Development:** Developing cross-platform native mobile applications using Flutter or React Native for convenient smartphone access.
4. **Multi-Language Localization:** Introducing support for regional Indian languages (such as Hindi, Malayalam, and Tamil) to make the marketplace widely accessible.
5. **AI-Powered Recommendation Engine:** Utilizing machine learning algorithms to offer personalized destination, package, and service suggestions based on user travel history.

9.3 Conclusion

The **Tourism Promotion Website** successfully bridges the gap between travelers and local communities by offering a centralized, transparent digital marketplace built with Node.js, Express.js, and MySQL. In direct alignment with **SDG 8: Decent Work and Economic Growth**, the platform democratizes digital access for micro-entrepreneurs, guides, and hospitality providers while offering tourists an authentic discovery and booking experience.

Key outcomes of the project include:

- **Economic Empowerment of Local Providers:** Delivered a dedicated, zero-cost digital portal enabling local businesses to market services directly to visitors without intermediary commission overheads.
- **Robust Full-Stack Implementation:** Implemented a modular, role-based architecture featuring secure JWT authentication, structured RESTful APIs, and an intuitive responsive user interface.

CHAPTER 10

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CHAPTER 11

APPENDIX

11.1 Sample Code

Application Entry Point (server.js)

```
require('dotenv').config();
const express  = require('express');
const cors     = require('cors');
const path     = require('path');
const session  = require('express-session');
const app      = express();

// Global Middleware Configuration
app.use(cors());
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(session({
    secret: process.env.SESSION_SECRET || 'tourism_session_secret_2026',
    resave: false,
    saveUninitialized: false,
    cookie: { secure: false, httpOnly: true, maxAge: 7 * 24 * 60 * 60 * 1000 }
}));

// Static Assets & Role-Based Dashboard Protection
app.use(express.static(path.join(__dirname, 'public')));
app.use((req, res, next) => {
    const url = req.originalUrl;
    const isDashboard = url.startsWith('/admin')
        || url.startsWith('/user')
        || url.startsWith('/provider');
    if (isDashboard) {
        if (!req.session || !req.session.user) return res.redirect('/login.html');
    }
    next();
});

// Core REST API Route Mounting
app.use('/api/auth',          require('./routes/auth'));
app.use('/api/destinations',  require('./routes/destinations'));
app.use('/api/bookings',      require('./routes/bookings'));
app.use('/api/services',      require('./routes/services'));
app.use('/api/packages',      require('./routes/packages'));
app.use('/api/payments',      require('./routes/payments'));
app.use('/api/analytics',     require('./routes/analytics'));

// Server Port Listener
```

```
42 const PORT = process.env.PORT || 3000;
43 app.listen(PORT, () => {
44     console.log('Tourism Promotion Marketplace running on port '
           + PORT);
45 });
```

Listing 11.1: server.js – Express.js Main Application Entry Point

Authentication Middleware (`middleware/auth.js`)

```
const jwt = require('jsonwebtoken');
const SECRET = process.env.JWT_SECRET || 'tourism_promo_secret_2026';

/**
 * Authentication and Role-Based Authorization Middleware
 * Validates JSON Web Tokens (JWT) and checks user permissions
 */
module.exports = function requireAuth(roles = []) {
    return (req, res, next) => {
        const header = req.headers.authorization || '';
        if (!header.startsWith('Bearer ')) {
            return res.status(401).json({
                success: false,
                error: 'Authentication token required.'
            });
        }

        const token = header.split(' ')[1];
        try {
            const payload = jwt.verify(token, SECRET);
            if (roles.length && !roles.includes(payload.role)) {
                return res.status(403).json({
                    success: false,
                    error: 'Access denied: insufficient permissions.'
                });
            }
            req.user = payload;
            next();
        } catch (err) {
            return res.status(401).json({
                success: false,
                error: 'Invalid or expired session token.'
            });
        }
    };
};
```

Listing 11.2: auth.js – JWT Authentication and Role Validation Middleware

11.2 Screenshots

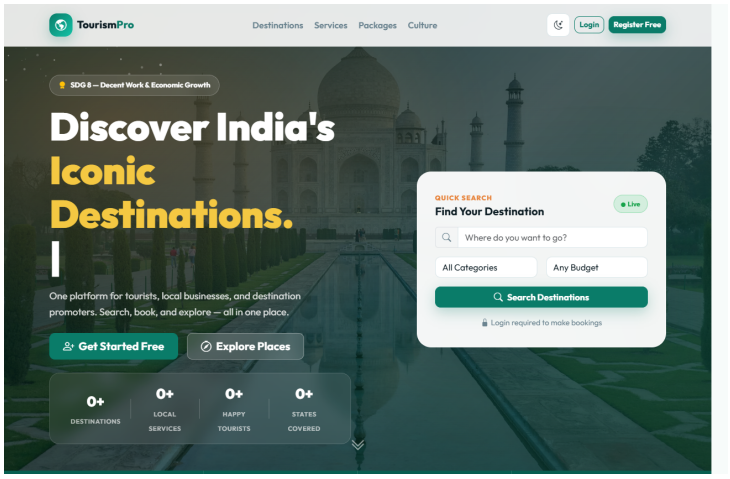


Figure 11.1: Landing Page – Tourism Promotion Website

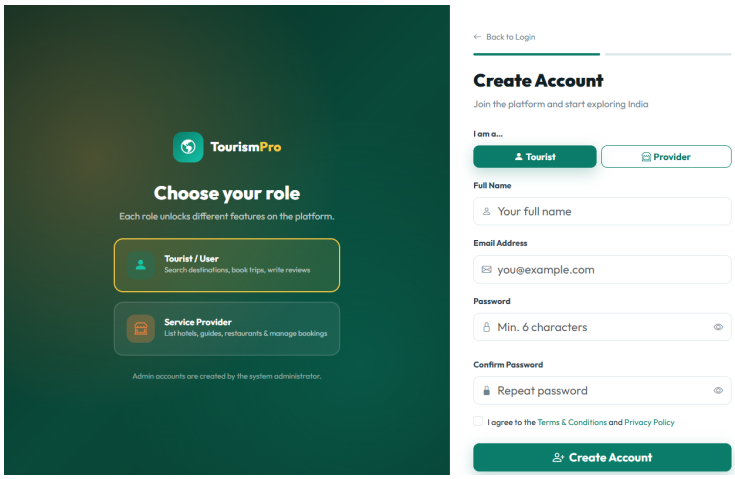


Figure 11.2: User Registration Page

11.3 Git Logs

The Git logs demonstrate the development history, version control activities, and continuous integration commits throughout the project lifecycle.

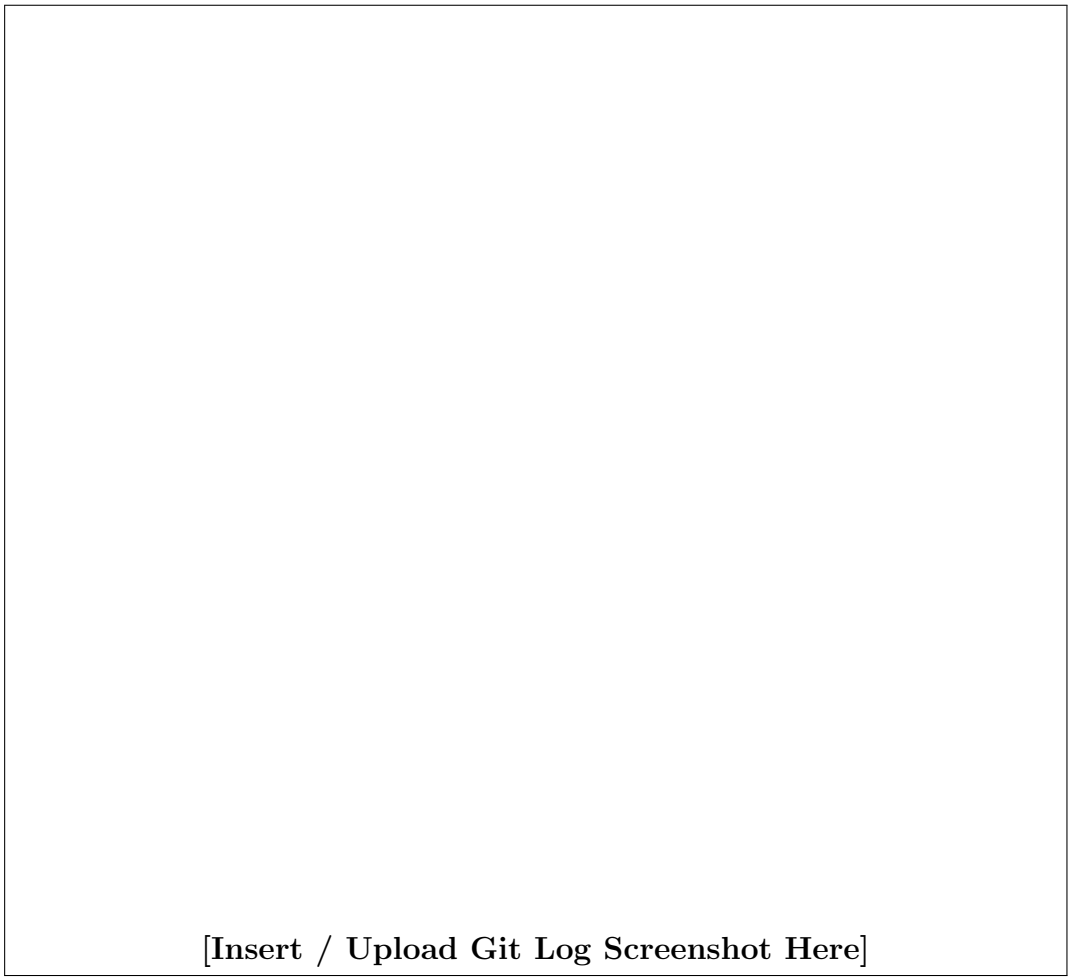


Figure 11.3: Git Log of the Project Development